Novus Robot – Quick start Guide

academic version

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Novus Hi-Tech Robotic Systemz Ltd.

Contents

1 Overview to Novus Robot

Novus is a state-of-the-art robotic system meticulously engineered to excel in durability, agility, and endurance. Designed to endure the rigors of continuous industrial operations, Novus features advanced high-performance batteries that ensure uninterrupted functionality, allowing it to work tirelessly alongside you in even the most demanding environments.

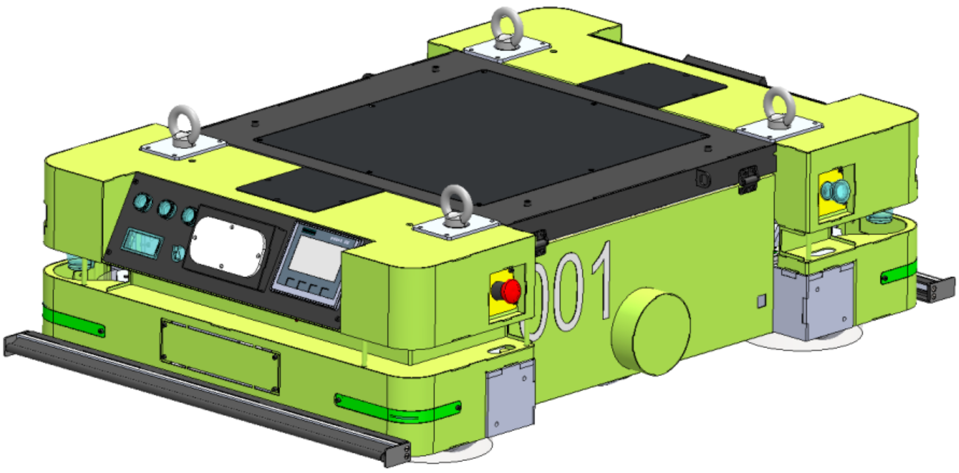


Figure 1: Novus Robot

Its robust construction and superior speed make it an invaluable asset in automating complex industrial tasks, enhancing both efficiency and productivity. Novus is equipped with the capacity to accommodate a wide array of additional components, including custom sensors, actuators, laptops, embedded boards, and more. This adaptability enables it to seamlessly integrate and operate with an extensive range of technological tools, effectively transforming itself into a highly intelligent and versatile work partner tailored to your specific needs.

Whether you require intricate data collection, precise control, or sophisticated automation, Novus is designed to handle the load with ease, making it a cornerstone of innovation in your industrial automation efforts.

2 Information's on this Manual

This manual is designed to make the equipment more accessible and the associated instructions clearer for operators and owners. It is essential that all personnel involved with the NOVUS Robot read and understand this manual before beginning any work. Adherence to all safety guidelines and instructions is crucial for ensuring a safe working environment.

The manual also provides instruction and information about,

1. A list of safety regulations and accident prevention.

2. The operating procedure of the Novus Robot.

> The latest version and additional documentation for specialized applications can be found on our support website. https://novushitech.com/

2.1 Safety Instructions

Safety precautions are highlighted with symbols and signals, each indicating the severity of potential hazards. It is crucial to follow these safety guidelines carefully to prevent accidents, personal injuries, and damage to property.

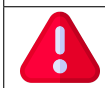
Symbol	Description
	**CAUTION:** Indicates a situation, which can lead to minor or moderate personal injury or property damage, if precautionary measures are ignored.
	**WARNING:** Indicates a situation, which can lead to severe personal injury, if the precautionary measures are ignored.
	**DANGER:** Indicates a situation, which can lead to death or severe personal injury, if the precautionary measures are ignored.

Table 1: Safety Symbols and Descriptions

Before operating the robot or performing any maintenance, it is essential that the operator reads and fully understands all cautionary and danger notes provided in the manual.

Additionally, the robot is equipped with tags that identify specific risks. These risks could potentially cause harm to both the operator and the robot if not properly addressed.

2.2 LED Indication

LED Status	Explanation
	RED SOLID: Error in robot operation. **RED BLINK:** Low Battery.
	GREEN SOLID: Robot is in AUTO Mode. **GREEN BLINK:** Robot is either in Manual mode or in Auto mode waiting for start button to be pressed.
	YELLOW SOLID: Robot stop zone has been breached. **YELLOW BLINK:** Robot's warning zone has been breached.

Table 2: Led lights and Descriptions

3 Important Hardware Components

3.1 Sensors

The robot is equipped with several preinstalled sensors designed to capture spatial information about its surroundings. It features a 3D LiDAR system that provides detailed point cloud data, offering a comprehensive view of the environment. Additionally, a depth camera is mounted at the front of the robot, supplying both depth measurements and visual imagery.

(a) LSLiDAR (b) Intel Real Sense Depth Camera

Figure 2: Preinstalled sensors on the Novus Robot

You can access the specification of these sensors from the below given links:
Robo Sense LiDAR Specifications: `https://www.lslidar.com/products/`
Depth Camera Specification: `https://www.intelrealsense.com/depth-camera-d457/`

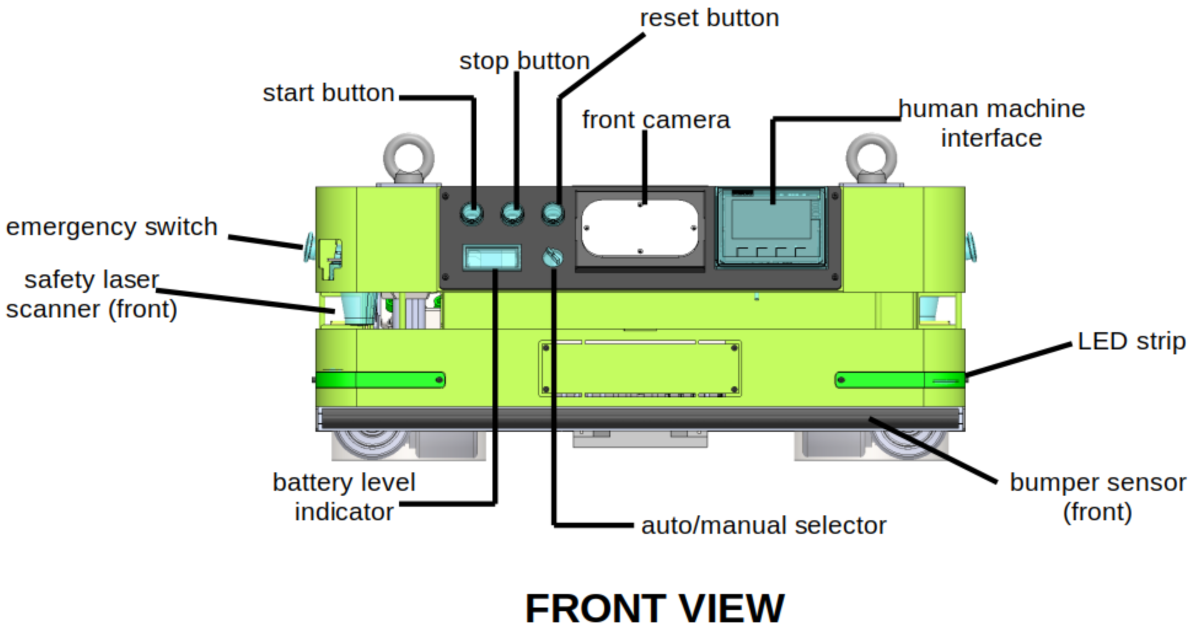


Figure 3: Front View of the Novus Robot

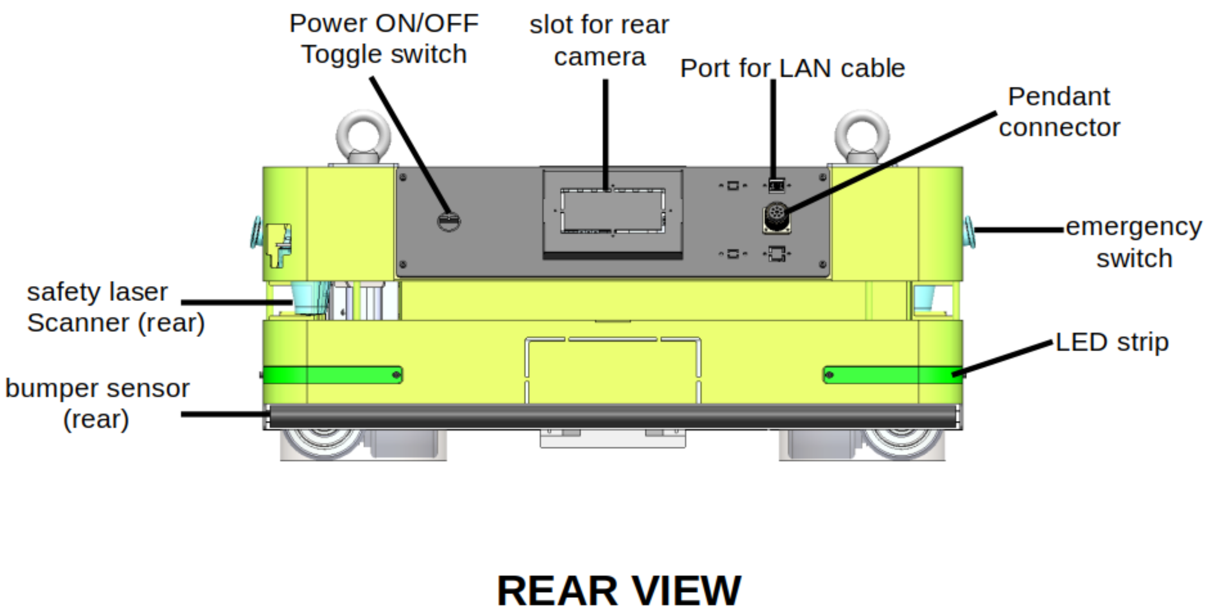


Figure 4: Rear View of the Novus Robot

3.2 Safety Components

For enhanced safety, the robot is equipped with two types of sensors: a contact-based **bumper sensor** and a non-contact safety **laser scanner**. The robot utilizes two diagonally mounted safety laser scanners to detect obstacles at the scan plane height. In addition, bumper sensors are installed at both the front and rear of the robot to provide a physical safety backup. These bumper sensors serve as a fail-safe in the event of a safety laser scanner malfunction and help detect obstacles that are lower than the safety scan height.

(a) Safety Laser Scanner

(b) Bumper sensor

Figure 5: Safety Sensors on the Novus Robot

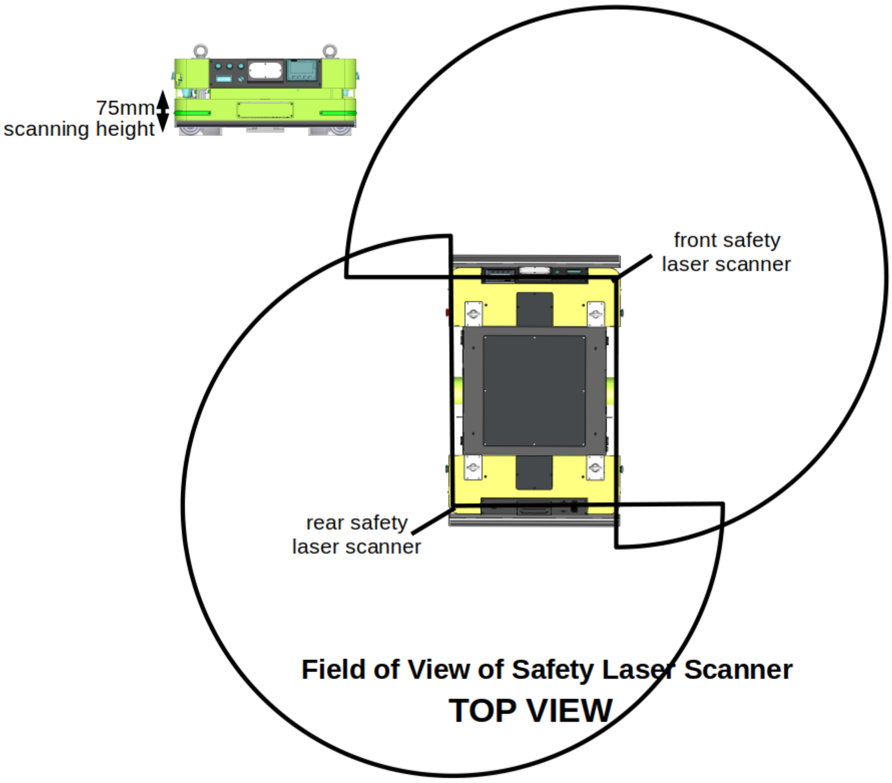


Figure 6: Field of View of Safety Laser Scanners and Scanning height

4 Basic Operations

> Safety Before Start:
>
> **Before starting the operation, it is absolutely necessary that the operator has read and understood all notes specified in this manual. Besides, the robot is provided also with tags for the identification of specific risks, which may cause damage to both the operator and robot if not duly taken into account.**

4.1 Operating Procedure

In order to start the Novus following checks must be done by the operator every time accesses the robot.

1. Do the visual inspection of the robot ensuring all the components are rigidly mounted.
2. Verify the battery voltage with the multi meter to be in excess of 25V.
3. Verify that the robot track is free from obstructions.
4. Verify that there are no loose cables hanging around in the robot.

Once these checks are done, the operator can connect the battery connector with the connector provided on the Novus robot and follow the process.

> **Usually the Battery can remain connected with the Novus robot all the times. Only in the extreme case when the Battery is completely discharged, it needs to be charged with the help of an external charger.**

4.2 Pre-Power On:

1. Make sure battery connectors are attched properly and covers are closed.
2. Make sure that the robot is in Manual Mode (toggle switch at the back of the robot).
3. Make sure that the Emergency Stop switches are released.

4.3 Power On:

1. Turn on the robot by toggling the Power ON/OFF switch. You will notice an LED pattern display on the LED strips present on the four corners of the robot, a boot up screen on the Human Machine Interface (HMI) and the battery level indicator will turn ON. If either of these does not happen, within 10 seconds, verify if the battery connections are proper.
2. After a few seconds, the system boot up processes will completed and the blue push button at the back of the robot would start blinking. Press and hold this button for 2 seconds twice to arm the motor controller.

3. The LED strip on the robot will turn GREEN once the robot is ready to move in AUTO mode.

> Spare Error: If you encounter this error, it indicates that the robot is currently in AUTO mode upon startup. To resolve this issue, simply turn the mode selector knob to the manual position.

4.4 Power Off:

1. Before powering off the system, ensure that the robot has come to a complete halt.
2. Toggle the power ON/OFF switch to OFF state. This will turn the robot off.

4.5 Mode of Operation:

- **Manual Mode:** Manual mode would be useful for off path maneuvering of Novus robot through Pendant.
- **Auto Mode:** Auto Mode would be the one where Novus robot will in self guided mode.

4.5.1 Operating in Manual Mode:

The following steps must be executed sequentially to move the Novus robot manually. The manual control of the robot is achieved via the pendant as shown below:

1. Switch ON the robot.
2. Set the robot to Manual Control using mode selection selector switch.
3. Reset robot using Reset Push Button.
4. Wait for 8 seconds, motor controller booting time.
5. To make the robot move forward, use **UP** push button provided on Pendant.
6. To make the robot move backward, use **DW** push button provided on Pendant.
7. To make the robot turn, use **L** and **R** push buttons provided on Pendant.

> In Manual mode, the Novus robot expect velocity commands from **/cmd_vel_manual** topic.

> In Manual Mode, the safety sensors are not active.

4.5.2 Operating in Auto Mode:

> Before operating in Auto mode make sure to configure the Remote PC to the Robot's IP_Address as mentioned in the section 5.1. And do the Master Slave Configuration as mentioned in the section 5.2

1. Switch ON the robot.

2. Set the robot to Auto mode using mode selection selector switch.
3. Reset robot using reset push button.
4. wait for 8 seconds, motor controller booting time.
5. To move the robot in the **Auto mode** you have to publish velocity command to the **/cmd_vel_stage** topic.

In Auto mode robot expect velocity commands from /**cmd_vel_stage** topic.

Follow the mentioned command to publish to the /**cmd_vel_stage** topic.

```
$ rostopic pub /cmd_vel_stage geometry_msgs/Twist "linear:
  x: 0.0
  y: 0.0
  z: 0.0
angular:
  x: 0.0
  y: 0.0
  z: 0.1"
```

Ensure that you start by publishing small velocity values. The velocities are expressed in meters per second (m/s).

6. Press Start Push Button.
7. Now you will see that robot starts to move.
8. Use emergency stop button to stop the Novus robot in any case of emergency.

When operating the robot in Auto mode, if its designated safety zone is breached, it will stop and remain stationary. To resume movement, you need to switch to Manual mode, then back to Auto mode , and finally press the Start button again.

After publishing the velocity command in AUTO mode, while pressing the start button you may breach the safety zone, preventing the robot from moving. To avoid this, set an appropriate safety zone based on your requirements.

We have some predefined safety zones. To utilize these zones, please follow the steps outlined below:

1. On HMI click START 》DIAGNOSTICS 》FRONT TIM
2. Here the default value will be NULL.
3. You can click on Field Selection and choose your desired safety zone field.

Based on our recommendations, Field 1 is the optimal choice for operating the robot in AUTO mode.
Similarly you can set the REAR TIM for the rear safety zone.

5 Software

> The following steps are designed for Ubuntu.

> Before following these steps make sure that you have ROS Noetic installed on your machine.

5.1 Steps to configure the PC to the Robot's IP_Address configuration:

Before connecting to the robot, configure your laptop/ development board to a network series having IP_address of **192.168.100.xxx** (except 192.168.100.104)

To configure the network to a **192.168.100.xxx** series follow the below given steps:

1. go to Settings > Network
2. click on the +.
3. then on Identity > Name: Here give a name to your network.
4. then go to IP_V4 then click on Manual.
5. Now on Address type your desired IP_Address: as 192.168.100.xxx and Netmask as255.255.255.0
6. At last click on Add

> When you are connecting your laptop/development board to the robot make sure that your wired connection is in this configuration.

5.2 Steps to do Master Slave Configuration:

> **Connect your laptop/development board to the Novus Robot using a LAN cable. Configure your laptop/development broad as the slave and Novus Robot as Master.**

Follow the below given steps to do the Master Salve configuration:

1. go to the **/etc** directory of the remote PC and add the below mentioned lines to the **hosts** file.

```
$ cd /etc
$ sudo nano hosts
```

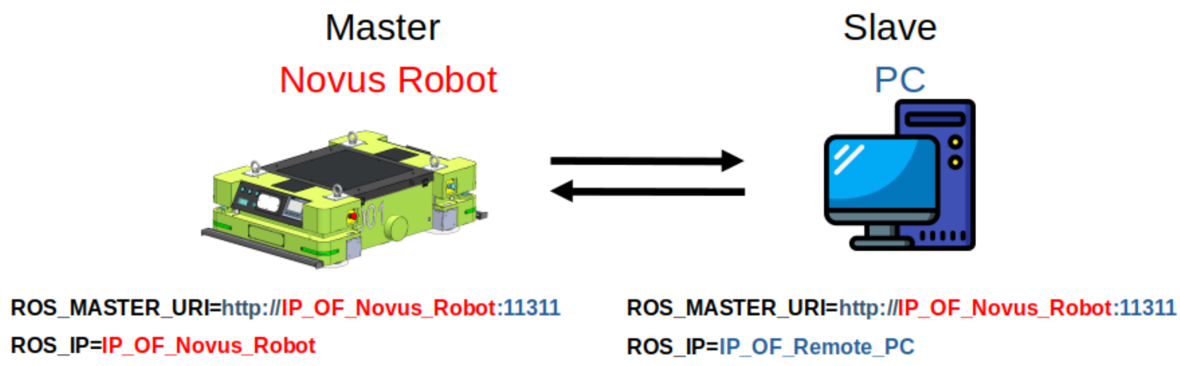


Figure 7: Master Slave Configuration

The hosts file will looks like this:

```
IP_addr_of_your_computer        localhost
IP_addr_of_your_computer        Your ComputerName
192.168.100.104                 nuc # Add these lines to your hosts file

# The following lines are desirable for IPv6 capable hosts
::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
```

IP_address of the robot = 192.168.100.104
Computer Name of the robot = nuc

2. After adding the above mentioned lines in the hosts file. Run the following commands in the terminal.

```
$ export ROS_MASTER_URI=http://192.168.100.104:11311/
```

This URI tells each node where to find the ROS Master.

```
$ export ROS_IP=192.168.100.xxx    # The IP_address of the PC.
```

The IP_address of the PC must be the one which is in the same series with the Novus Robot's IP_address series.

To ensure that other node can properly discover and communicate with this node, you should also set the **ROS_HOSTNAME** or **ROS_IP** variable to the IP_address of the slave machine.

5.3 Steps to access the Sensor Data and Actuators

Once you do the Master Slave Configuration, now you can see the topics that are being published by Novus Robot.

Run the following to list the topics.

```
$ rostopic list
```

Among all the listed topics the important and useful topics are:

- /camera_front/color/camera_info
- /camera_front/color/image_raw
- /camera_front/depth/camera_info
- /camera_front/depth/color/points
- /cmd_vel_manual
- /cmd_vel_stage
- /odom
- /tf
- /tf_static
- /velodyne_points

To access the LiDAR data you can subscribe to the topic **/velodyne_points.**

```
$ rostopic echo /velodyne_points
```

To access the camera data you can subscribe to the topic **/camera_front/color/image_raw**.

```
$ rostopic echo /camera_front/color/image_raw
```

To see the status of the robot's input output, you can subscribe to the topic **/robot_io_states**

```
$ rostopic echo /robot_io_states
```

This will return the states of the input output switches like start button, stop button , emergency switch etc. You can have a look at it by doing a echo first.

To make the robot move you can publish to the topic **/cmd_vel_manual.**

```
$ rostopic pub /cmd_vel_manual geometry_msgs/Twist "linear:
  x: 0.0
  y: 0.0
  z: 0.0
angular:
  x: 0.0
  y: 0.0
  z: 0.1"
```

If the robot is in **Manual state** then to make the robot move either you have to use **/cmd_vel_manual** topic or move the robot using **Pendant**. To move the robot using pendant follow the procedure mentioned in the section 4.5.1

If the robot is in **Auto state** then to make the robot move you have to use **/cmd_vel_stage** topic and follow the procedure mentioned in the section 4.5.2

6 Robot Specifications

Dimensions

Length	1100 mm
Width	750 mm
Height	410 mm
Ground Clearance	35 mm
Weight (Without load)	250 kg
Load Surface	600 mm x 830 mm
Wheel Diameter	Drive wheel: 210 mm Caster Wheel: 100 mm

Payload

Robot payload	500 kg

Speed and performance

Battery Running time	8 hrs
Max speed	forward: 1.25 m/s backward: 1.25 m/s
Acceleration	0.2 m/s^2 (full payload, flat surface)
De-Accelartion	0.2 m/s^2 (full payload, flat surface)

Power

Battery	LiFPo 24V, 80 Ah
Charging Time	up to 4 hours (0-80%: 3 hrs)
External Charger	Input: 200-240 V AC, 50-60 Hz Output: 24V
Battery Charging Cycle	Minimum 1500 cycles